Fiscal Year:	FY 2023	**Task Last Updated:**	FY 01/10/2023
PI Name:	Stankovic, Aleksandra Ph.D.		
Project Title:	Modeling Individual and Team Problem-Solving Skills for the Establishment of Training Metrics, Standards, and Guidelines for Long-Duration Space Missions		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Charlestown	**State:**	MA
Zip Code:	02129-2020	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2019 HERO 80JSC018N0001-HHCHFBP: Human Health Countermeasures, Human Factors, Behavioral Performance. Appendix D
Start Date:	03/09/2020	**End Date:**	03/08/2024
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 03/08/2024 per A. Beitman/JSC (Ed., 2/6/23)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Robinson, Stephen Ph.D. (University of California, Davis) Strangman, Gary (Massachusetts General Hospital/Harvard Medical School) Caldwell, Barrett (Purdue University)		
Grant/Contract No.:	80NSSC20K0844		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Motivation: Future human lunar exploration will necessitate that astronaut crews work autonomously to solve unexpected and time-critical problems, and will demand increased cooperation between humans and robotic/autonomous systems. While the ability of these multi-agent teams to adapt to unforeseen and off-nominal situations will be integral to mission success and safety, frameworks and recommendations for cultivating team-based problem-solving are currently lacking. Goals: This project aims to characterize team problem-solving strategies for future long duration space exploration tasks, specifically in the context of Lunar and Cis-Lunar operations. This work will generate findings that will help shape team-based concepts of operations and functional requirements, will contribute towards the development of a set of validated measures and metrics for assessing team problem-solving, and will produce an approach to modeling dynamic team problem-solving performance over time, with the goal of helping determine training effectiveness. Outcomes of this study will also help shape future decision and execution support systems. This project has 3 integrated specific aims: Aim 1: Through literature review and in collaboration with teaming subject matter experts, we will (a) identify key teaming and problem-solving knowledge gaps and attributes of interest, and will (b) develop an approach for the in-situ assessment and characterization of individual and team-based problem solving using operationally-deployable metrics and measures. (Completed.) Aim 2: Through data collection in Earth-bound analogs for Lunar/Cis-Lunar activities, we will examine attributes contributing to individual and team problem-solving effectiveness, and will characterize how such skills are developed and used across a variety of operational tasks, under both normal and emergency conditions. (Currently in progress.) Aim 3: Leveraging the insights gained in Aims 1 & 2, we will develop a set of team-based concepts of operations and functional requirements, which may ultimately be used to inform future training guidelines and recommendations for individual and team problem-solving, including for the effective interface with on-board support systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We are undertaking an operationally-relevant assessment of team problem-solving skills in the context of upcoming Cis-Lunar exploration, with the ultimate goal of contributing towards the establishment of future autonomous team-based concepts of operations and functional design requirements for system development, operations, and training.
Task Progress:	Project accomplishments to date include the following items: completion of definition-phase exit milestones (e.g., identification and recruitment of teaming expert co-Investigator, identification of subject matter experts for informational interviewing and data collection, updating of the project timeline and definition of next phase milestones, completion of the Life Sciences Data Archive/LSDA data sharing agreement); completion of preliminary literature review (published Jan 2023 in the journal Human Factors); definition and refinement of experimental methodology and research protocol; definition of field site requirements, identification of potential field sites, and early-stage logistical planning for field research; and drafting of Institutional Review Board (IRB) materials. We are continuing to coordinate with the Element and NASA Research Operations and Integration (ROI) for the long-duration deployment of our research to an extreme analog site (South Pole Station for winter-over 2023) and are currently in the process of logistical planning and coordination for over-winter data collection. We are collaborating with field site partners to coordinate additional in-field data collection with domain-specific expert teams working in the areas of science return, search and rescue/emergency response, and remote medical care. In the next 6 months, we anticipate further extensive collection of field-based data across several extreme and remote analog research sites and progression into the work of Research Aim 3.
Bibliography Type:	Description: (Last Updated: 03/18/2026)
Abstracts for Journals and Proceedings	Harris K*, Stankovic A*, Thoolen S, Strangman G, Caldwell B, and Robinson S. "Evaluating team performance and problem-solving skills for lunar construction: A review of team dynamics challenges in long-duration spaceflight and complex construction scenarios." Human Factors Journal: Special Issue on Human Factors and Ergonomics in Spaceflight. *Equal contributions as joint first authors. Journal Article. , Jan-2023
Abstracts for Journals and Proceedings	Thoolen S, Harris K, Caldwell B, Strangman G, Robinson S, and Stankovic A. "Team problem solving for Lunar exploration." NASA Human Research Program Investigator's Workshop. Conference Abstract. , Feb-2022
Abstracts for Journals and Proceedings	Harris, K., Porter, A., Haberman, M., Stankovic, A. "Automation: A help or hinderance in spaceflight and remote medical settings?" 1st International Conference of Aerospace Medicine (ICAM). Conference Abstract. , Sep-2022
Articles in Peer-reviewed Journals	Harris KM, Stankovic A, Thoolen S, Strangman G, Caldwell B, Robinson SK. "Team dynamics and collaborative problem-solving for Lunar construction: Lessons from complex construction scenarios on Earth." Hum Factors. 2023 Jan 6;00187208221148610. https://doi.org/10.1177/00187208221148610 ; PubMed PMID: 36607842 , Jan-2023